

JavaScript Functions - Complete Guide

What are Functions?

Functions are reusable blocks of code that perform specific tasks. They help organize code, make it DRY (Don't Repeat Yourself), and are essential for JavaScript programming.

1. Function Declaration

```
function functionName(parameter1, parameter2) {  
  // code here  
  return result;  
}
```

2. Function Expression

```
const functionName = function(param) {  
  return param * 2;  
};
```

3. Arrow Function (Modern)

```
const functionName = (param) => {  
  return param * 2;  
};  
  
// Or shorter:  
const double = (x) => x * 2;
```

Real Examples:

Example 1: Calculate Grade

```
function getGrade(marks) {  
  if (marks >= 90) return 'A+';  
  if (marks >= 80) return 'A';  
  if (marks >= 70) return 'B';  
  return 'C';  
}
```

```
console.log(getGrade(85)); // Output: A
```

Example 2: Calculate Bill

```
const calculateBill = (amount) => {  
  const tax = amount * 0.17;  
  return amount + tax;  
};  
  
console.log(calculateBill(1000)); // 1170
```

Key Differences:

- Function Declaration: Can be called before definition (hoisting)
- Arrow Function: Cannot use 'this' keyword (advanced topic)
- All return values, all accept parameters